Taylor Daniels

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ABOUT

Math PhD with expertise in Asymptotic Analysis, and significant background and experience in Spectral Geometry, Mathematical Physics, and programming. Currently learning Data Science and Machine Learning—my background is helping me make rapid progress. Looking for roles in Applied Math, Machine Learning, Simulation, and Data Science.

EDUCATION

Ph.D., **Mathematics**, Purdue Univ., West Lafayette, IN **GPA**: 3.70 **2015 – 2024**

- Miscellaneous Scholarship, Purdue Mathematics 2021 – 2022
- Excellence in Teaching Award, Purdue Mathematics Fall 2020

M.S., **Mathematics**, Univ. of Louisville, Louisville, KY **GPA**: 4.00 **2014 – 2015**

B.S., **Mathematics**, Univ. of Louisville, Louisville, KY **GPA**: 3.85 **2009 – 2014**

- Bickel Scholarship, UofL Mathematics 2013 – 2014
- Pedigo Scholarship, UofL Mathematics 2012 – 2013
- Trustee's Scholarship, UofL 2009 – 2014
- Humana Scholarship, Humana Inc., Louisville, KY 2009 – 2012

SKILLS

Programming and Computation: Python, numpy, scipy, Sage, Mathematica, matplotlib, Git, JSON
ML/DS: Scikit-learn, pandas, data collection and cleaning, text parsing, regression and classification
Mathematics: Analytic Number Theory, Integer Partitions, Spectral Geometry, Hyperbolic Geometry
Physics: Gauge Theory, Quantum Mechanics, Lagrangian/Hamiltonian mechanics
Certifications: Summer 2026 Data Science Boot Camp – The Erdős Institute
Soft Skills: Excellent ability to communicate technical topics; excellent written and verbal communication
Languages: English – (Native); **Chinese/Mandarin** – (Intermediate); **Spanish** – (Intermediate)

EXPERIENCE

Visiting Assistant Professor, Purdue Univ. Mathematics **2024 – 2026**

- Lead instructor for 8 sections of 200-level Differential Eqns. and Linear Algebra

Graduate Teaching/Research Assistant, Purdue Univ. Mathematics **2015 – 2024**

- Instructed & graded courses ranging from Calculus up to graduate-level Complex Analysis

Instructor, STARTALK Summer Program, Louisville, KY **Summer 2015**

- Fully planned and taught 4 weeks of 6-hour lessons for a course in Mandarin Chinese for students grade 6-8

Teaching Assistant, Univ. of Louisville Mathematics **2012 – 2015**

- Tutored and assisted teaching work for undergraduate courses from Algebra to Calculus

SELECTED PROJECTS

Erdős Institute Certificate in Data Science: **Summer 2026**

- Analyzed fair balance and statistical biases in randomly generated Pokémon battles.
- Implemented Javascript tools to reverse-engineer random team generation.
- Developed Python tools to analyze 18,000+ Pokémon Showdown battle logs.
- Skills used: **Python**, **Javascript**, **pandas**, **Scikit-learn**, **Data Science**, **Machine Learning**

Doctoral and Professional Research: **2021 –**

- Computed and analyzed long-term asymptotic behaviors of 1000's of novel weighted integer-partition sequences.
- Developed parallel computation algorithms to overcome (sub-)exponential search space; computational speed increased 10-fold.
- Skills used: **Python**, **GP/Pari**, **Sage**, **remote computation**

Domain coloring: **2022 – 2024**

- Adapted Python library 'cplot' to produce high-resolution magnitude-phase function plots for collaborators.
- The generated function plots guided theoretical math research, and featured in several academic publications.
- Skills used: **matplotlib**, **cplot**, **Mathematica**

Graduate-level Mathematical Physics: **2017 – 2019**

- 18 months of advanced-level study of Gauge Theory and the Vector Bundle framework of physical fields.
- Graduate Quantum Mechanics course at Purdue; independent study Lagrangian/Hamiltonian Mechanics.
- Skills used: **Gauge Theory**, **Quantum Mechanics**, **Topology**, **Differential Geometry**

PUBLICATIONS

Vanishing Coefficients in Products of Quintuple Products, T. Daniels et. al; submitted; arXiv **2026**
The p-dissection of a product of quintuple products, T. Daniels et. al.; submitted; arXiv **2026**
Vanishings of the Legendre-17 signed partition numbers, T. Daniels; arXiv **2026**
Biasymptotics of the Möbius- and Liouville-signed partition numbers, T. Daniels, submitted; arXiv **–**

Legendre-signed partition numbers, T. Daniels, Published in J. Math. Anal. Appl. **542** doi	**2025**
Vanishing coefficients in two q-*series…*, T. Daniels; Published in Res. Number Theory **10**; doi	**2024**
The Asymptotics of Some Signed Partition Numbers (Doctoral Thesis), T. Daniels; doi	**2024**
Bounds on the Möbius-signed partition numbers, T. Daniels; Published in Ramanujan J. **65;** doi	**2024**
Differential Properties of the HFE *Cryptosystem*, T. Daniels et. al.; Published by PQCrypto 2014;	**2014**

ORGANIZATIONS

Vice President, Math Club, Univ. of Louisville	**2014 – 2015**
Math Club, Univ. of Louisville	**2012 – 2014**
Chinese Learner's Society, Univ. of Louisville	**2009 – 2012**
President, Chinese Learner's Society, Univ. of Louisville	**2010 – 2011**